Dataset Description

IIT-H Campus Life — Mess Duration Prediction

What is this dataset?

This is a **synthetic** dataset that simulates how long a student spends eating in the mess at IIT Hyderabad. Each row represents one meal visit by one student on one day.

Target variable: `mess_duration_min` — time spent in the mess (in minutes, range 10–120).

Features

Feature	**Type**	**Description**
`student_id`	int	Unique student ID (2101001–2101199)
`day_of_week`	int	0 = Monday … 6 = Sunday
`meal_time`	int	0 = Breakfast, 1 = Lunch, 2 = Dinner
`meal_type`	int	0 = Veg, 1 = Non-veg
`is_weekend`	int	1 if Saturday or Sunday, else 0
`class_day`	int	0 = has classes today, 1 = free day
`assignment_deadline`	int	0 = deadline today, 1 = no deadline
`temperature_c`	float	Temperature in °C (15–42)
`humidity`	float	Relative humidity % (0–100)
`wind_speed`	float	Wind speed in m/s (0–25)
`rain_mm`	float	Rainfall in mm (0–60)
`air_quality_index`	float	AQI — higher means worse air (20–350)
`rising_time_min`	int	Wake-up time in minutes from midnight (240–660)
`sleeping_time_min`	int	Sleep time in minutes from midnight (1140–1439)
`mess_duration_min`	float	**TARGET** — time spent in mess

How was the data generated?

Step 1 — Schedule Features

500 records were created spread randomly over a 15-day window starting on a Monday. For each record we assigned a random meal time (breakfast / lunch / dinner), meal type (≈28% chance of non-veg), and whether the day had classes (85% chance on weekdays, 10% on weekends). Three days in the window were randomly picked as assignment deadline days.

Step 2 — Weather

Daily weather (temperature, humidity, wind, rain, AQI) was generated using smooth sine/cosine curves to mimic Hyderabad's climate, plus random Gaussian noise on top. Each record also gets a tiny extra noise to simulate variation within the same day. Rain chance is linked to humidity — higher humidity means a higher probability of rain, which is realistic.

Step 3 — Student Profiles

Each student was given a personal habitual wake-up time (≈7:30 AM on average), sleep time (≈11:00 PM), and eating pace (some eat faster, some slower). These are sampled randomly but realistically. Per record, the wake-up and sleep times shift based on context:

- Weekends → wake up ≈28 min later, sleep ≈18 min later
- Deadline day → wake up ≈8 min earlier, sleep ≈25 min later

Step 4 — Target Generation (`mess_duration_min`)

The time spent in the mess is built using an **additive formula**:

Component	**Effect**
Base duration (breakfast / lunch / dinner)	18 / 34 / 44 min
Student eating pace	personal offset (± few min)
Weekend	+4.5 min
Non-veg meal	+2.5 min
Free day (no class)	+3.5 min
Deadline today	−3.0 min
High humidity	slightly longer
Rain	+0.20× rainfall mm
Extreme temperature	slightly longer
Bad AQI	slightly shorter
Late riser at breakfast	eats faster
Random noise	$\pm \sim 3.8$ min

Final values are clipped to [10, 120] minutes.

Augmentation (500 → 5000 Samples)

The 500-sample base dataset is repeated 10 times. Then small Gaussian noise is added **only** to the continuous input features (weather + wake/sleep times) to simulate measurement variation. All categorical columns and the target are kept exactly as-is.

Feature	Noise added (std)
`temperature_c`	$\pm 0.7\,°\mathrm{C}$
`humidity`	$\pm 2.5\,\%$
`wind_speed`	$\pm 0.5\,\mathrm{m/s}$
`rain_mm`	$\pm 1.2\,\mathrm{mm}$
`air_quality_index`	± 7.0
`rising_time_min`	$\pm 9\,\mathrm{min}$
`sleeping_time_min`	$\pm 9\,\mathrm{min}$

Dataset Sizes

Dataset	Rows	File
Base	500	`iith_campus_life_500.csv`
Augmented	5000	`iith_campus_life_5000.csv`

Both files share the same 15 columns. All results are fully reproducible using `--seed 7`.